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Liquid crystal nanodroplets in solution

W. Michael Brown,^{a)} Matt K. Petersen,^{b)} Steven J. Plimpton,^{c)} and Gary S. Grest^{d)}
Sandia National Laboratories, Albuquerque, New Mexico 87185, USA

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The aggregation of liquid crystal nanodroplets from a homogeneous solution is studied by molecular dynamics simulations. The liquid crystal particles are modeled as elongated ellipsoidal Gay–Berne particles while the solvent is modeled as spherical Lennard–Jones particles. Extending previous studies of Berardi *et al.* [J. Chem. Phys. **126**, 044905 (2007)], we find that liquid crystal nanodroplets are not stable and that after sufficiently long times the nanodroplets always aggregate into a single large droplet. Results describing the droplet shape and orientation for different temperatures and shear rates are presented. The implementation of the Gay–Berne potential for biaxial ellipsoidal particles in a parallel molecular dynamics code is also briefly discussed. © 2009 American Institute of Physics. [DOI: [10.1063/1.3058435](https://doi.org/10.1063/1.3058435)]

I. INTRODUCTION

Nematic droplets of liquid crystals have long been studied as electro-optical devices with applications including optical switches, light shutters, and reflective displays.^{1–3} While these materials typically involve micron-sized droplets embedded in a solid polymer matrix, nematic emulsions consisting of droplets in a continuous liquid solvent have also been employed.^{1–4} Because the physical properties of dispersed liquid crystal droplets depend on the phase and structural characteristics of the droplets,^{2,4} understanding the crystallization/self-assembly process is important for targeted design of new materials. An important consideration in the use of nematic emulsions for electro-optical applications is the fact that the emulsions are only metastable. In most cases the liquid crystal nanodroplets are stabilized by cross-linking the solution monomers by photopolymerization⁵ or cooling to form a glassy matrix.^{6,7}

In recent work, Berardi *et al.*⁸ simulated aggregation of liquid crystal nanodroplets from a homogeneous binary solution. In this study the liquid crystal particles were modeled as elongated ellipsoidal Gay–Berne⁹ particles while the solvent is modeled as spherical Lennard–Jones (LJ) particles. The simulations of Berardi *et al.* suggested that liquid crystal nanodroplets could be stable in a solvent for a range of temperatures and concentrations. For the system they studied which consisted of ellipsoidal particles with a 3:1 aspect ratio in a solvent of spherical particles, their simulations showed that nanodroplets of a few hundred mesogens were stable. While most of the simulations were for systems of 5000 total particles, one run on a system of 40 000 particles suggested that the nanodroplets with a characteristic size of a few hundred mesogens was stable, at least for the length of the simulations Berardi *et al.*⁸ were able to study. In their study they carefully characterized the influence of solute-

solvent interaction, temperature, and composition on the morphology, structure, anisotropy, and size of the resulting droplets.

From a computational standpoint, determining the long term stability of nanodroplets is difficult to establish due to limitations on the time scales that can be achieved with molecular dynamics and the expense of anisotropic pair potentials required for modeling aspherical particles. In order to address these issues, we have modified the LAMMPS molecular dynamics package¹⁰ to facilitate simulation of aspherical particles using the Gay–Berne⁹ and other aspherical potentials. This has allowed us to extend the studies of Berardi *et al.* to longer time scales for the large system of 40 000 particles. We find that the liquid crystal nanodroplets are only metastable and after long times coalesce into a single large droplet. To verify that a single nanodroplet is the most stable state of the mixture, a number of simulations at different temperatures were initiated from a single large spherical liquid crystalline droplet. The droplets quickly changed shape to that of the aggregate formed from a homogeneous solution and were stable for the length of the simulation run. In addition to characterizing the shape and orientation of the droplets in equilibrium, we also studied the effects of shear stress on the structure of the droplet to mimic rheological response induced by manufacturing processes, e.g., extrusion of a thin film.

In Sec. II we present details of the simulation model and methodology. In Sec. III, we present results for aggregation from a homogeneous solution and stability of single large droplets. Results for the effect of shear on the shape and orientation of the droplet are also presented. In Sec. IV, we present a brief summary and suggestions for future work. In the Appendix we give implementation details of the aspherical particle models.

II. MODEL AND METHODOLOGY

The Gay–Berne potential is a single-site interaction potential for rigid molecules derived from heuristic modifications to a Gaussian overlap potential.⁹ The potential, which

^{a)}Electronic mail: wmbrown@sandia.gov.

^{b)}Electronic mail: mkpeter@sandia.gov.

^{c)}Electronic mail: sjplimp@sandia.gov.

^{d)}Electronic mail: gsgrest@sandia.gov.

can be considered as an anisotropic and shifted LJ 6-12 interaction, has been extensively used for the modeling of mesogenic systems. Although it was originally presented as a soft potential for ellipsoidal particles of equivalent size, it has since been generalized for dissimilar biaxial ellipsoids.¹¹ Following the notation in Everaers and Ejtehadi,¹² the potential can be written as a product of three terms,

$$U = U_r \cdot \eta \cdot \chi, \quad (1)$$

parametrized by the ellipsoid shapes and relative interaction energies. For shape, the ellipsoid semiaxes a_i , b_i , and c_i for each particle i are specified to form the diagonal elements of a “shape” matrix, $\mathbf{S}_i = \text{diag}(a_i, b_i, c_i)$. Likewise, the relative well depths ϵ_{ai} , ϵ_{bi} , and ϵ_{ci} for particles interacting along the corresponding semiaxes (side-to-side, face-to-face, and end-to-end interactions) give the matrix $\mathbf{E}_i = \text{diag}(\epsilon_{ai}, \epsilon_{bi}, \epsilon_{ci})$. The orientation of each particle is given here by the rotation matrix $\mathbf{A}_i$ representing the transformation from the laboratory frame to the body frame.

In Eq. (1), U_r represents the shifted LJ interaction given by the interparticle distance h , the atomic interaction radius σ , and the shift factor γ ,

$$U_r = 4\epsilon(\varrho^{12} - \varrho^6), \quad (2)$$

$$\varrho = \frac{\sigma}{h + \gamma\sigma}. \quad (3)$$

Because the particles are aspherical, the interparticle distance h is not between particle centers but rather represents the distance of closest approach between particles. While an exact calculation of h is nontrivial,¹³ an approximation has been given by Perram *et al.*¹⁴ that is commonly used in Gay-Berne calculations,

$$h = r - \left[\frac{1}{2} \hat{\mathbf{r}}^T \mathbf{G}^{-1} \hat{\mathbf{r}} \right]^{-1/2}, \quad (4)$$

where $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ is the particle center separation, $r = |\mathbf{r}|$ is the center-to-center distance, $\hat{\mathbf{r}} = \mathbf{r}/r$, and

$$\mathbf{G} = \mathbf{A}_1^T \mathbf{S}_1^2 \mathbf{A}_1 + \mathbf{A}_2^T \mathbf{S}_2^2 \mathbf{A}_2. \quad (5)$$

In addition to the distance of closest approach, the interaction anisotropy is characterized by the distance-independent terms η and χ that control interaction strength based on the particle shapes and relative well depths, respectively,

$$\eta = \left[\frac{2s_1 s_2}{\det(\mathbf{G})} \right]^{v/2}, \quad (6)$$

$$s_i = [a_i b_i + c_i c_i][a_i b_i]^{1/2}, \quad (7)$$

and

$$\chi = [2\hat{\mathbf{r}}^T \mathbf{B}^{-1} \hat{\mathbf{r}}]^\mu, \quad (8)$$

$$\mathbf{B} = \mathbf{A}_1^T \mathbf{E}_1^2 \mathbf{A}_1 + \mathbf{A}_2^T \mathbf{E}_2^2 \mathbf{A}_2. \quad (9)$$

The parameters μ and v in Eqs. (6) and (8) are empirically determined exponents that can be tuned to adjust the potential.

The analytic expressions for the forces and torques as well as details of the parallel implementation of the Gay-

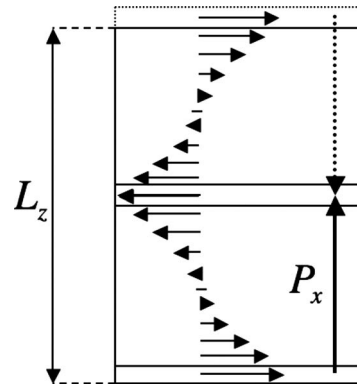


FIG. 1. Velocity profile for Müller-Plathe reverse perturbation method.

Berne potential for biaxial ellipsoidal particles in our LAMMPS MD code^{10,15} are given in the Appendix.

We have used the same model parameters as Berardi *et al.*⁸ These parameters are described in dimensionless units in terms of the characteristic length σ_0 , energy ϵ_0 , and mass m_0 . The solute is modeled as a uniaxial prolate ellipsoid with a mass $1.5m_0$, an aspect ratio of 3, and Gay-Berne parameters $\epsilon_{\text{meso}}^{\text{meso}} = \epsilon_0$, $\sigma_{\text{meso}}^{\text{meso}} = \sigma_0$, $a_i^{\text{meso}} = b_i^{\text{meso}} = \sigma_0$, $c_i^{\text{meso}} = 3\sigma_0$, $\epsilon_a^{\text{meso}} = \epsilon_b^{\text{meso}} = \epsilon_0$, and $\epsilon_c^{\text{meso}} = 0.2\epsilon_0$. For the solvent, the mass is m_0 and the Gay-Berne parameters $\epsilon^{\text{solv}} = \epsilon_0$, $\sigma^{\text{solv}} = \sigma_0$, $a_i^{\text{solv}} = b_i^{\text{solv}} = c_i^{\text{solv}} = \sigma_0$, and $\epsilon_a^{\text{solv}} = \epsilon_b^{\text{solv}} = \epsilon_c^{\text{solv}} = 3.0\epsilon_0$. For the solute-solvent interactions, $\epsilon^{\text{cross}} = 2.0\epsilon_0$. In all cases, the Gay-Berne model parameters have been set as $\gamma = 1$, $\mu = 1$, and $v = 3$. A cutoff radius for the potential of $r_c = 4\sigma_0$ and a neighbor list radius of $4.8\sigma_0$ were used.

The equilibrium molecular dynamics simulations were carried out in an isothermal-isobaric (*NPT*) ensemble with a time step of 0.002τ ($\tau = \sigma_0[m_0/\epsilon_0]^{1/2}$) and pressure $P = 8.0\epsilon_0/\sigma_0^3$. The damping parameters for the thermostat and barostat were both set to 0.5τ . The temperature $T^* = k_B T/\epsilon_0$ of the simulations varied from 2.2 to 3.9. A mole fraction of 0.04 prolate ellipsoids was studied with a total of approximately 40 000 particles.

In order to investigate the response of the large aggregate to a shearing flow, we subjected the droplet to varying shear rates using the Müller-Plathe¹⁶ (MP) reverse perturbation method. The MP method induces a shear flow by imposing a momentum flux. The periodic box is divided into a number of slabs along one coordinate, for example, the z direction (see Fig. 1). Particles in the top slab are inspected to find the particle with the largest momentum orthogonal (p_x) to the slab normal, while particles in the middle slab are inspected to find the particle with the smallest (most negative) momentum. In effect, these are the particles moving most counter to the desired streaming direction. Once this particle pair is identified, the momentum between the two is exchanged, and the process is repeated a specified number of time steps later. The shear rate is modulated by adjusting the frequency of the momentum swaps, by swapping between pairs with less than the extremal momenta, or both. In these simulations of binary systems, the momenta of only solvent particles are exchanged.

As can be seen in Fig. 1, the induced velocity profile is composed of two approximately linear profiles, mirrored

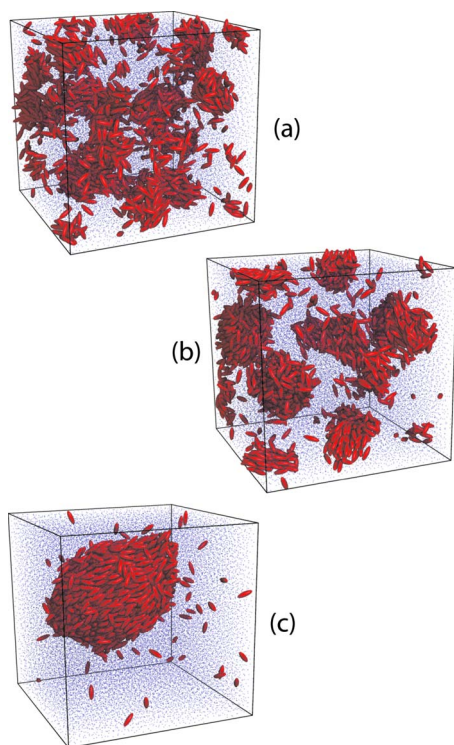


FIG. 2. (Color online) Snapshots showing solute aggregation at $t=440\tau$ (a), $t=880\tau$ (b), and $t=7040\tau$ (c) for $T^*=2.4$. Solvent is rendered at 10% size for clarity.

about the middle slab and periodic boundary. To keep the aggregate within the Couette flow region, we constructed the sheared system by doubling the system along the z direction. The resulting starting configuration was composed of two identical droplets; each contained entirely within the Couette flow region. Reduced shear rates, $(\partial v_x / \partial z)^* = (\partial v_x / \partial z) \times (m\sigma^2 / \epsilon)^{1/2}$, from 1×10^{-2} to $5 \times 10^{-5} \tau^{-1}$ were simulated. The shear simulations were run at constant volume. To maintain the temperature the atoms were weakly coupled to the Langevin thermostat with a damping constant $\Gamma = 0.01 \tau^{-1}$ in the vorticity (y) direction. The shearing runs ranged in length from $t = 2.4 \times 10^4 \tau$ to $10^5 \tau$. These comparatively long runs were necessary to allow for the establishment of the shear flow and potential aggregate restructuring relative to the flow. For each shear rate the first $10^4 \tau$ were discarded before properties of the droplet were determined.

III. RESULTS

For our initial simulations, we followed the procedure described by Berardi *et al.*⁸ starting with a 5×10^3 time step equilibration of a uniform homogeneous solution with $\epsilon^{\text{cross}} = 3.0\epsilon_0$ to avoid demixing. In these simulations, a total of 42 875 particles were used (with 1720 ellipsoid mesogens). This pre-equilibration was utilized to perform similar simulations, but for much longer time scales than previously reported. This is important because the size of the multiple, distinct nanodroplets formed from solute aggregation as reported in the previous study is influenced by the length of the simulation. For example, for temperature $T^*=2.4$ and simulation time $t=880\tau$, we observe that distinct nanodroplets

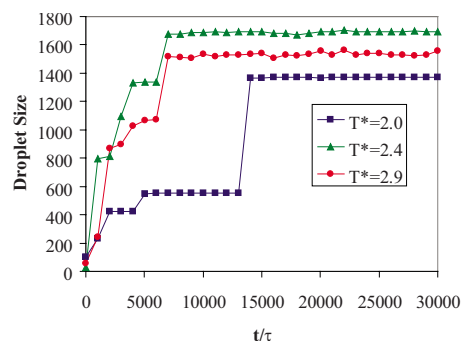


FIG. 3. (Color online) Growth of the largest droplet for three temperatures.

have formed with similar size to those reported by Berardi *et al.* However, by $t=6700\tau$, a single aggregate has formed (Fig. 2).

The dynamics for the growth process are shown in Fig. 3 for three temperatures. In this figure the size of the largest droplet at a given time is reported according to Stillinger's definition¹⁷ such that a solute particle is included in a droplet if the contact distance [approximated by Eq. (4)] is within $0.5\sigma_0$ of another particle in the droplet. For $T^*=2.9$ and $T^*=2.4$, a single aggregate is formed by $t=6700\tau$. The size of the final aggregate is somewhat larger at $T^*=2.4$ than at $T^*=2.9$ as there are fewer isolated mesogens in solution. We would expect the aggregate at $T^*=2.0$ to be largest, however, solute demixing results in near-frozen solvent at this temperature. After $t=15\,000\tau$, the two remaining solute droplets move very little within the solvent. The discontinuous jumps in size observed in Fig. 3 occur from the merging of smaller distinct droplets that are concurrently formed. Aside from fluctuations in number of droplets that occur when droplets merge due to the contact distance threshold, an increase in the number of droplets is not seen during any of the simulations.

As an alternative approach to investigating the stability of the aggregate, we performed simulations with a starting configuration of a spherical aggregate of 1636 solute particles in solvent (consisting of 39 088 particles). With this starting configuration, we ran simulations for times exceeding $30\,000\tau$ for a range of temperatures. Results for the size of the largest droplet are shown in Fig. 4 for six temperatures. For $T^* \leq 3.5$ a single, large nanodroplet is stable with

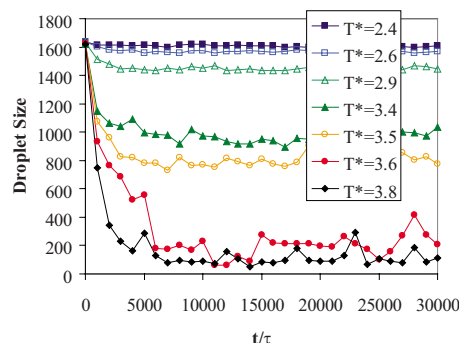


FIG. 4. (Color online) Size of the largest droplet as a function of time for an initial spherical droplet of size 1636. The results are shown for $T^*=2.4, 2.6, 2.9, 3.4, 3.5, 3.6$, and 3.8 (from top to bottom).

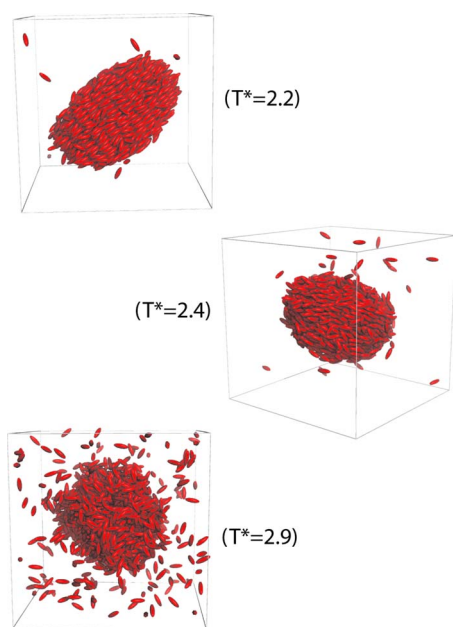


FIG. 5. (Color online) Snapshots showing aggregate shape at different temperatures ($t=30\,000\tau$). For clarity, solvent is not shown and the configuration has been translated so that the aggregates are approximately centered.

increasing amounts of mesogens dispersed in the solvent. For higher temperature the droplet breaks up into many smaller droplets. These results suggest that the phase boundary for the isotropic to uniform mixture is shifted to slightly higher temperature than reported by Berardi *et al.*⁸ In Fig. 5 snapshots of the droplet at late times are shown for three temperatures. In no case did the aggregates break apart into multiple droplets, however, changes in the size and mesophase of the aggregate were observed as illustrated in Fig. 5.

The formation of elongated prolate nanodroplets has been observed in these simulations at short time scales and in experiment.⁸ We have observed similar results in the aggregates studied here with the formation of prolate structures at lower temperatures. The shape of the largest droplet can be described using the ratio E_r of the square roots of the largest and smallest eigenvalues obtained from diagonalization of the aggregate inertia tensor. This gives the aspect ratio from the largest and smallest orthogonal axes within the droplet. This metric is appropriate for the aggregation of uniaxial particles where the ratio between at least two of the axes should be 1 on average. When $E_r=1$, the aggregate is symmetric. For the simulations studied, the aggregate aspect ratios all converged by 7000τ . The average ratios, $\langle E_r \rangle$, from simulation times $20\,000\tau$ to $30\,000\tau$ are shown in Fig. 6. The largest droplet becomes more elongated as the simulation temperature decreases. At a temperature $T^*=2.9$, the aggregate is nearly spherical (Fig. 5) while at $T^*=2.2$ it has an average aspect ratio of nearly 2 to 1.

As one measure of the mesophase ordering, Berardi *et al.* examined the instantaneous orientational order parameter P_2 . $P_2=1/2n\sum_i(3\cos^2\beta_i-1)$ for each particle i in the aggregate with angle β_i with the aggregate director. The results are shown in Fig. 7 for different temperatures and are similar to those reported previously.⁸ Taking the average of P_2 over the final third of each simulation, we observe a decrease in the

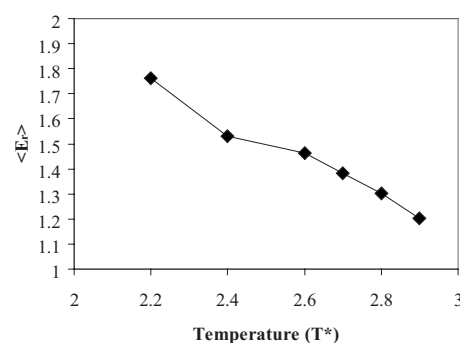


FIG. 6. Average shape $\langle E_r \rangle$ of the largest droplet at different temperatures from $t=20\,000\tau$ to $t=30\,000\tau$.

orientational ordering of the solute with a isotropic aggregate forming for higher temperature. The results for E_r and P_2 obtained from the simulations starting from a single spherical aggregate are comparable to those calculated for late time for the aggregate obtained from a homogeneous solution.

As can be seen in Fig. 5., the aggregates begin to develop smectic ordering at the lower temperatures investigated. Some quantitative measure of this effect can be obtained using the translational order parameter, $\rho_{\parallel}=1/n\sum_i\cos(2\pi/lr_{\parallel})$, where $r_{\parallel}$ is the position of the center of particle i along the aggregate director (with aggregate center of mass at the origin) and l is the layer spacing. Taking the mean value $\langle \rho_{\parallel} \rangle$ of the order parameter from $t=20\,000\tau$ to $t=30\,000\tau$ we find that there is little translational order at $T^*>2.4$ ($\langle \rho_{\parallel} \rangle=0.002$). At $T^*=2.2$, the mean translational order is 0.53 and at $T^*=2.0$, $\langle \rho_{\parallel} \rangle=0.70$. Because the mesogen

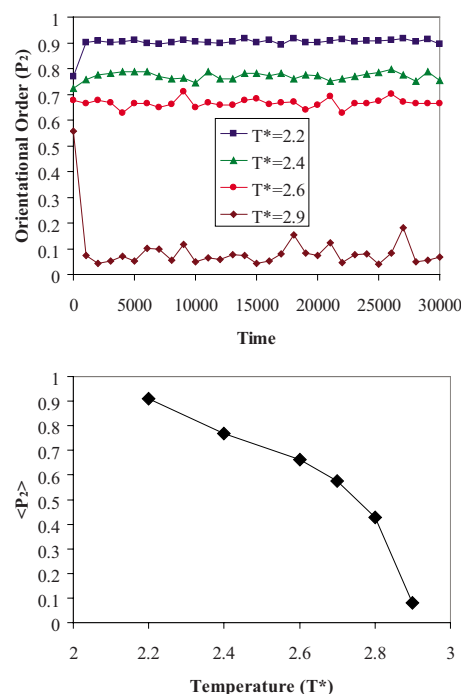


FIG. 7. (Color online) Orientational order of the largest droplet as a function of temperatures. (Upper) Time evolution of the orientational order parameter. Each data point represents an instantaneous value of P_2 at a given time step. (Lower) Mean value $\langle P_2 \rangle$ of the orientational order parameter for each droplet. The average is taken over the instantaneous value of P_2 for every time step from $t=20\,000\tau$ to $t=30\,000\tau$.

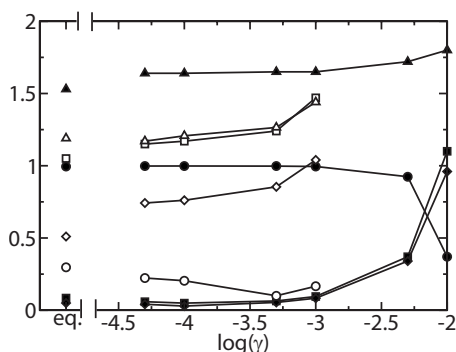


FIG. 8. Droplet shape and the relationship between the long axis of the droplet to the aggregate director for $T^*=2.4$ for shear rates between 1×10^{-2} to $5 \times 10^{-3} \tau^{-1}$ and at equilibrium (eq). Results are for aspect ratio (triangles), cosine of the angle between the droplet long axis and director (circles), difference between the long axis and director unit vectors (squares), and the difference of the x component only of the long axis and director vectors (diamonds).

simulations were performed in solvent, $\rho_{||}$ as calculated here can be an underestimate due to deviations in the number and structure of particles from an ideal symmetry about the plane normal to the droplet director (the aggregate center of mass does not necessarily lie exactly in the middle of a layer). The results are comparable, however, with the previous report of a nematic-smectic phase boundary at $T^*=2.0$.⁸

To study the response of the droplet to shear we investigated a nematic droplet at $T^*=2.4$ and an isotropic droplet at $T^*=2.9$. A droplet was chosen whose aggregate directors were oriented roughly perpendicular to the shear direction. The aggregate orientational ordering P_2 of each droplet remained unchanged from the equilibrium structure for both temperatures for all shear rates investigated. Likewise, the orientation of the droplet's aggregate director D was unaltered by the shear flow. However, for large shear rates, the long axis L of the nematic droplet did reorient to align more closely with the shear direction. Figure 8 quantifies this non-intuitive effect. The directions of L and D are nominally equivalent at low shear rates (circles in plot). Above a shear rate of $10^{-3} \tau^{-1}$ the two vectors diverge and the angle between them becomes appreciable. Thus instead of the entire drop rotating as a rigid body to align more nearly with the flow, only the shape of the drop does so (L); the individual ellipsoidal particles retain their original orientation (D). This difference can be seen in Fig. 9 which is a representative configuration for the nematic droplet under shear where L has rotated to align more closely with V_x , but D has not. This configuration was stable for the $10^5 \tau$ time scale of the shear simulation. Note, however, that this effect was not seen for the isotropic droplet at $T^*=2.9$. As in the equilibrium simulations, a nearly spherical isotropic aggregate persisted, exhibiting little correlation between D and L , i.e., the two vectors were not aligned even at low shear rates.

Figure 8 also plots the total difference (squares) between the D and L vectors and the component of difference in the shear direction x (diamonds). The x -direction component accounts for the majority of the total difference at the largest shear rates for the nematic droplet. As the shear rate decreases, the difference in the x components contributes a less

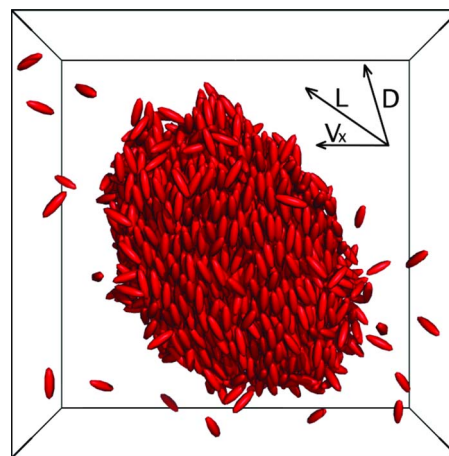


FIG. 9. (Color online) A representative configuration for a nematic droplet at $T^*=2.4$ under a shear rate of $5 \times 10^{-3} \tau^{-1}$. The shear direction, V_x , droplet director, D , and long axis, L , are labeled in the figure inset.

significant portion. At a shear rate of $10^{-4} \tau^{-1}$, the difference in the x components contributes no more than the others to the total difference. In the isotropic case, the difference in these two vectors is large at all shear rates, while the x component constitutes no more of the difference than the others. The difference in the two vectors is what would be expected on average for randomly oriented vectors.

Finally, the shape of both the nematic and isotropic droplets was affected by the shear flow. The aspect ratio (triangles in Fig. 8) of the nematic droplet increased by $\approx 10\%$ between the equilibrium value and the largest shear rate. The isotropic droplet aspect ratio increase was more dramatic, 20% from the equilibrium value, presumably due to the relatively more labile constituent ellipsoids. For incrementally larger shear rates than those presented in Fig. 8, the droplets eventually distort enough to join across the periodic boundary. These trajectories, which occurred at significantly lower shear rates for the isotropic system, were excluded from the analysis.

IV. DISCUSSION

The aggregation of liquid crystal nanodroplets has been studied using molecular dynamics simulations. By implementing the Gay-Berne potential⁹ for biaxial ellipsoidal particles in a parallel MD code we were able to extend the previous studies of Berardi *et al.*⁸ to large systems for longer times. We find that small liquid crystal nanodroplets are not stable in solution and that after sufficiently long times the nanodroplets always aggregate into a single large droplet. The stability of the larger droplets was confirmed by starting the simulations from a single large droplet. For temperatures in the nematic and isotropic regimes the single droplet was stable over the length of the run and only broke up for higher temperatures in the uniform phase. All of the results were for a solute mole fraction of 0.04 but we expect similar results for other mole fractions. The shape and orientation of the liquid crystal droplets was also determined and found to agree with previous simulations.⁸ The effect of shearing the droplets was found to be minimal except at very high shear rates for both nematic and isotropic droplets.

As observed experimentally, nanodroplets are in most cases stabilized by cross-linking the solution or cooling to form a glassy matrix. Both of these routes to obtain liquid crystal nanodroplets can be explored numerically. For the former one needs to simultaneously cross-link the solvent atoms while the solute and solvent phase separate. For the latter the temperature can be reduced at a controlled rate so that the solvent freezes, halting the phase separation. Some evidence for this is seen in our simulations for $T^*=2.0$ where the aggregation was halted as the solvent atoms froze. To avoid effects of the solvent crystallizing, the solvent could be modeled as short bead-spring polymer chains. One important question to be addressed in both of these simulations is how to control the size distribution of the resulting nanodroplets. With the performance increase offered by parallel machines, simulations of this type are presently feasible and will lead to new insight into this important class of new materials.

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APPENDIX A: IMPLEMENTATION DETAILS

1. Analytic forces and torques

Due to the complexity of the expressions necessary for calculation of forces and torques in aspherical particles, finite difference methods are often employed at the potential sacrifice of accuracy and efficiency.¹⁸ Discussion on the general derivation of expressions for intermolecular forces and torques have been described¹⁹ and analytic expressions for the biaxial Gay–Berne potential have been given recently.²⁰ In summary, the force can be calculated as

$$\mathbf{f} = -\eta \left(U_r \cdot \frac{\partial \chi}{\partial \mathbf{r}} + \chi \cdot \frac{\partial U_r}{\partial \mathbf{r}} \right), \quad (\text{A1})$$

where

$$\frac{\partial U_r}{\partial \mathbf{r}} = \alpha \hat{\mathbf{r}} + r^{-2} \frac{\alpha(r-h)^3}{2} [\kappa - (\kappa^T \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}}] \quad (\text{A2})$$

and

$$\frac{\partial \chi}{\partial \mathbf{r}} = -r^{-2} \cdot 4.0 \cdot [\iota - (\iota^T \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}}] \cdot \boldsymbol{\mu} \cdot \chi^{(\mu-1)/\mu}, \quad (\text{A3})$$

with $\alpha = 24\epsilon(2\varrho^{13} - \varrho^7)/\sigma$, $\kappa = \mathbf{G}^{-1} \cdot \mathbf{r}$, and $\iota = \mathbf{B}^{-1} \cdot \mathbf{r}$.

The torque for each particle i is given by

$$\tau_i = U_r \eta \frac{\partial \chi}{\partial \mathbf{q}_i} + \chi \left(U_r \frac{\partial \eta}{\partial \mathbf{q}_i} + \eta \frac{\partial U_r}{\partial \mathbf{q}_i} \right), \quad (\text{A4})$$

where

$$\frac{\partial U_r}{\partial \mathbf{q}_i} = \mathbf{A}_i \cdot \left(-\kappa^T \cdot \mathbf{A}_i^T \mathbf{S}_i^2 \mathbf{A}_i \times \left[\frac{\alpha r^{-2} \kappa}{-2} (r-h)^3 \right] \right), \quad (\text{A5})$$

and

$$\frac{\partial \chi}{\partial \mathbf{q}_i} = 4.0 \cdot r^{-2} \cdot \mathbf{A}_i (-\iota^T \cdot \mathbf{A}_i^T \mathbf{E}_i^2 \mathbf{A}_i \times \iota). \quad (\text{A6})$$

It is not straightforward to write a matrix expression for $\partial \eta / \partial \mathbf{q}_i$ due to presence of the determinant. Instead, let $\mathbf{a}_{mi}$ be row m of the rotation matrix $\mathbf{A}_i$. Then,

$$\frac{\partial \eta}{\partial \mathbf{q}_i} = \mathbf{A}_i \cdot \sum_m \mathbf{a}_{mi} \times \mathbf{z}_{mi}, \quad (\text{A7})$$

where $\mathbf{z}_{mi}$ represents the row m of the matrix $\mathbf{Z}_i$ given by

$$\mathbf{Z}_i = -\frac{1}{2} \cdot \left[\frac{2s_1 s_2}{\det(\mathbf{G})} \right]^{v^2} \cdot \frac{v}{\det(\mathbf{G})} \cdot \mathbf{V}, \quad (\text{A8})$$

with $\mathbf{V} = [v_{my}]$ and $v_{my} = \det(\mathbf{G}) \cdot \text{trace}[\mathbf{G}^{-1} \cdot (\hat{\mathbf{p}}_y \otimes \mathbf{a}_m + \mathbf{a}_m \otimes \hat{\mathbf{p}}_y) \cdot \mathbf{s}_{mm}^2]$. Here, $\hat{\mathbf{p}}_y$ is the unit vector for the axes in the laboratory frame ($\hat{\mathbf{p}}_1 = [1, 0, 0]$, $\hat{\mathbf{p}}_2 = [0, 1, 0]$, and $\hat{\mathbf{p}}_3 = [0, 0, 1]$) and s_{mm} gives the m_{th} radius of the ellipsoid i .

2. Molecular dynamics

In the LAMMPS molecular dynamics code, simulations are parallelized via spatial-decomposition techniques that partition the simulation domain into smaller subdomains, one per processor. This typically achieves high parallel efficiencies so long as there are a few hundred particles per processor, which in the context of this work enables larger and much longer simulations of droplet systems to be performed.

An aspherical particle style was added to LAMMPS, suitable for ellipsoids, wherein each particle stores a quaternion orientation four-vector, as well as angular velocity, angular momentum, and torque vectors. In a parallel context, several of these attributes need to be communicated for ghost particles near subdomain boundaries to enable computation of pairwise properties. The biaxial Gay–Berne potential, with calculation of analytic forces and torques, was implemented which detects when a cheaper ellipsoid/sphere interaction can be substituted, as well as a much cheaper LJ 12-6 interaction when two spheres interact. These are important optimizations for the heterogeneous droplet simulations discussed here, since only a small fraction of pairwise interactions are between two ellipsoids.

Quaternion-based methods for time integration of ellipsoidal particles and thermostating/barostating of their rotational degrees of freedom were also added to the code. These avoid the well-known singularity issues that arise with rotational equations of motion formulated using Euler angles. The current version of LAMMPS (available for download from the WWW site) includes these various features and can run models of aspherical particles in the microcanonical (NVE) ensemble, the canonical (NVT) ensemble, or the isothermal-isobaric (NPT) ensemble with Berendsen or Nose–Hoover thermostats/barostats.

In comparison to a standard 12-6 potential, the computational cost of a Gay–Berne interaction between two ellipsoidal particles is approximately $20\times$ more expensive in our implementation. In serial, the homogeneous mixture simulations of 40 000 particles described here run at a rate of

approximately 9.2τ per hour on a 3.6 GHz Intel EM64T processor. The same system runs at 841.7τ per hour on 128 identical processors connected by an Infiniband network, allowing for an approximate speedup of $92\times$ in this case.

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